Technical Document #655: Cybersecurity - Comprehensive Overview

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Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Encryption converts plaintext into ciphertext to protect data confidentiality.
- Intrusion detection systems (IDS) monitor network traffic for suspicious activity.
- Multi-factor authentication (MFA) requires multiple forms of verification.